Forthcoming in Maria Ioannidou and Deni Mantzari (eds) *Research Handbook on Competition Law and Data Privacy* (Elgar 2023)



Exclusionary and Exploitative Abuse of Consumer Data

By Anca Daniela Chirita*

Abstract: This chapter advances a novel hybrid concept of abuse of consumer data, alongside algorithmic price discrimination and consumer data combinations, which have negative exclusionary and exploitative welfare effects on the fundamental freedom of business action and consumer choice, respectively. It advances legal interpretations to close the gap in the abuse of a dominant position when applied in conjunction with the Digital Markets Act (DMA) provisions relevant to consumer data combinations, digital tying, and algorithmic discrimination. This chapter argues that digital conglomerate mergers and killer acquisitions help sustain unfair consumer data–driven competitive advantage and consumer data combinations detrimental to as-efficient competitors of digital monopolists and consumers. It identifies legislative lacunae concerning the use of AI software algorithms for greater predictive analytics of consumer behaviour due to the availability of consumer experience, location, medical and financial data, and marketing strategies, such as personalised offers. For consumer justice, this chapter argues for economic consumer privacy inclusiveness, namely greater consumer autonomy over economic choices and preferences and greater consumer bargaining of reservation prices with further consumer empowerment in physical and virtual locations.

Keywords: exclusionary and exploitative consumer data abuse, digital monopolies, digital conglomerate mergers and acquisitions, algorithmic discrimination, consumer data combinations

Introduction

This chapter explores the emergence of a novel concept of consumer data abuse as a standalone form of abuse under Article 102 of the Treaty on the Functioning of the European Union (TFEU) in conjunction with relevant provisions of the Digital Markets Act. Hence, this chapter will reflect on the relevant case law on data abuse, including the latest statutory enactment, to advance novel forms of consumer data abuse, especially data combinations

Electronic copy available at: https://ssrn.com/abstract=4424967

and algorithmic price discrimination. This chapter aims to offer greater predictability to the dynamic nature of the anti-competitive conduct of digital monopolists working with rapidly expanding technological developments. Technological competition deviates from traditional competition[1] to the extent that innovation can consolidate entrenched digital monopolies.

As innovative technologies emerge in digital markets, new regulatory enactments could easily become redundant before implementation. Therefore, two options are carefully balanced. First, finding previous guiding principles, fundamental rights and freedoms, which could apply to digital market cases. Second, assessing the usefulness of prescriptive provisions for digital monopolists regarding the traditional concept of abuse; this is done to ensure greater clarity and legal certainty with respect to the concept of consumer data abuse.

1. Concept of Consumer Data Abuse

Over decades, the concept of abuse has amalgamated a diverse range of anti-competitive conduct by dominant undertakings. The original mandate of Article 102 was not to close the door to potential opportunities, which could capture novel forms of abusive conduct. Nevertheless, this expansion opened the prospect of dealing with an infinite column of unique categorisations of abuse of a dominant position, rather than securing the experience of a definitive catalogue of similar cases under a set of unifying guiding principles or fundamental rights or freedoms for both businesses and consumers. Over the years, judicial formalism has prevailed by supporting a conservative approach towards the competition goals, which seek to prevent market distortions for 'the detriment of the public interest, individual undertakings and consumers' with a notable addition of consumer 'well-being'.[2] Thus, the outcomes of individual cases have not signalled a major enforcement change.

A common feature of the common law and the continental codification systems was to judicially create or statutorily enact either a precedent grounded in significant judicial experience or to prove a legal provision offering certainty to a given position adopted in previous cases. Technological competition has massively disrupted the regulatory prospects and the judicial experience, making consumer data abuse difficult to predict due to legal

* Durham University Law School, Dr. iur. (Germany), Assistant Professor of Competition Law, Durham Institute for Data Science, UK. I would like to thank Maria Ioannidou and Deni Mantzari and the attendees of the University College London conference on 'Competition Law and Policy in a Data-Driven Economy' held in April 2023.

[1] See Mordecai Kurz, *The Market Power of Technology: Understanding the Second Gilded Age* (Columbia University Press 2023) 60.

[2] See Case T-399/16 *CK Telecoms UK Investments Ltd v EC* [2020] ECLI:EU:T:2020:217, para 93.

Electronic copy available at: https://ssrn.com/abstract=4424967

loopholes, such as zero-priced products or services, predictive algorithms, and unregulated marketing strategies. For multiple reasons, which are beyond the scope of this chapter, it has not been possible to structure a given set of prior cases into a unifying statutory regulation on the abuse of a dominant position. This would be a welcome, yet challenging, task for the future.

Thus, the enactment of the Digital Markets Act (hereinafter DMA)[3] offers momentum to contemplate how to accomplish this mission in the near future and reflect on whether this regulatory prospect could materialise a more detailed drafting of Article 102 as opposed to traditional law-making and judicial interpretation by the EU Courts of the concept of abuse. For the latter, the mainstream position[4] has favoured two major categorisations of abuse of monopoly power, which can be exclusionary of other businesses and/or exploitative of consumers. The hybrid development of an anti-competitive, exclusionary foreclosure, leading to consumer harm, attempted to unite the above categorisations into a concept of abuse that could, concomitantly, harm both businesses and consumers. The latest category of abuse hoped to reconcile opinion differences to meet business demands for market openness[5] and opportunity equality and deliver greater choice for the consumers. In addition, it hoped to consistently apply the same economic harm standard by referencing an exclusionary foreclosure of rivals and exploitative consumer harm.

[3] See Regulation (EU) 2022/1925 on contestable and fair markets in the digital sector and amending Directives (EU) 2019/1937 and 2020/1828 (Digital Markets Act, hereinafter DMA) [2022] OJ L 265/1.

[4] See Robert O'Donoghue and Jorge Padilla, *The Law and Economics of Article 102 TFEU* (Hart Publishing 2020) 294, opting for several categories of abuse, namely exploitative, exclusionary, discriminatory, tying and leveraging abuses; Bellamy & Child, *European Union of Competition* (Oxford University Press 2013) 791, para 10064, maintaining a dual classification of exclusionary and exploitative abuses; Miguel de la Mano, Renato Nazzini, and Hans Zenger, 'Article 102' in Jonathan Faull & Ali Nikpay (eds), *The EU Law of Competition* (Oxford University Press 2014) 388, para 4262, favouring a dualistic classical classification of exploitative, including discrimination, and exclusionary abuses; Thomas Eilmansberger, 'Articles 82-86 EC' in Günther Hirsch, Frank Montag, and Franz Jürgen Säcker (eds) *Competition Law: European Community Practice and Procedure* (Sweet & Maxwell 2008) 1117, paras 215170 and 215227, opting for a not so clearly cut distinction between exclusionary, including market-structure, and exploitative abuses, including the exploitation of a business partner; Richard Whish & David Bailey, *Competition Law* (Oxford University Press 2021) 209, opting for a tripartite classification of exploitative, exclusionary, and single market abuses; Alison Jones, Brenda Sufrin and Niamh Dunne*, Jones & Sufrin's EU Competition Law: Text, Cases, and Materials* (Oxford University Press 2019) 362, favouring four categories of exclusionary, exploitative, discriminatory and market integration-oriented abuses; Ioannis Lianos, Valentine Korah with Paolo Siciliani, *Competition Law: Analysis, Cases, & Materials* (Oxford University Press 2019) 1144, opting for a non-categorical classification including inter alia exclusionary, exploitative, and internal market abuses.

[5] On the value of openness with free competition, see Anca Daniela Chiriţă, 'Undistorted, (Un)fair Competition, Consumer Welfare and the Interpretation of Article 102 TFEU' (2010) 33 World Comp L & Econ R 420.

Electronic copy available at: https://ssrn.com/abstract=4424967

It is important to understand how this hybrid classification applies to a novel category of consumer data abuse, replacing the dualistic approach to exclusionary and exploitative effects. It is possible that a dominant undertaking exploits the predictive analytics tools of its business partners through aggregated data sharing or selling. Therefore, exploitative consumer data abuse includes intermediate consumers. Second, if the consumers are marginalised to the point of exclusion through an excessive loss of privacy, then an exclusionary abuse does not target only the as-efficient competitors of digital monopolists. Third, a more extensive classification of abuse has been an imperfect indicator of a clear materialisation of abuse with a common denominator of economic harm to businesses, consumers, or both. Such extensive catalogues of abuse rely on the development of a series of cases over time and an external form of abuse; however, without reflecting on the balance of justice for the consumers, the final judicial outcome, and the actual standard of economic harm. The shortcoming of an extensive abuse catalogue lies in dealing with isolated forms of abuse, which may never occur again. Nevertheless, there is the perception of a dense categorisation of abuse without the additional support of repeated judicial experience.

Therefore, this chapter aims to effectively align the digital markets cases about the prohibition of abuse of dominance as subordinate to an overarching concept of consumer data[6] abuse, using an analytical tripartite categorisation of exclusionary, exploitative, and hybrid consumer data abuses.

This chapter considers the DMA as a supplementary alternative to the traditional methods of competition law since digital markets represent an EU top enforcement priority.[7] Nonetheless, in several ways, the DMA signals its departure from the traditional concept to

[6] See, on competition law and big data, Maurice E Stucke, *Breaking Away: How to Regain Control Over Our Data, Privacy, and Autonomy* (Oxford University Press 2022); on the lack of equality and fairness regarding price discrimination, Ariel Ezrachi and Maurice E Stucke, *Virtual Competition: The Promise and Perils of the Algorithm-Driven Economy* (Harvard University Press 2016) 123; on free goods and price-centric antitrust, Maurice E Stucke and Allen P Grunes, *Big Data and Competition Policy* (Oxford University Press 2016) 107; reflecting on the inherent tensions among lawyers and economists regarding the optimal balance between the economic considerations of privacy and ethics, Chris Jay Hoofnagle, 'The Federal Trade Commission's Inner Privacy Struggle' in Evan Selinger, Jules Polonetsky and Omer Tene (eds) *The Cambridge Handbook of Consumer Privacy* (Cambridge University Press 2018) 173; OECD, *Consumer Data and Competition: A New Balancing Act for Online Markets?* (Going Digital Toolkit Policy Note, 2021); Drew Fudenberg and J Miguel Villas-Boas, 'Price Discrimination in the Digital Economy' in Martin Peitz and Joel Waldfogel (eds) *The Oxford Handbook of the Digital Economy* (Oxford University Press 2012) 255-69, specifically on how online price discrimination based on past purchasing history reveals valuable consumer data in sectors such as telecommunications, media, and banking.

[7] See the mandate of the EU President, 'A Europe Fit for the Digital Age' in EC, *Report on Competition Policy 2021* (2022) 337, 14.

Electronic copy available at: https://ssrn.com/abstract=4424967

lower the threshold of anti-trust intervention against unfair practices of not-yet-dominant gatekeeper undertakings.[8] The DMA's flexible adaption to include new materialisations of competitive unfairness by gatekeeper platforms aligns with the open-textured nature of Article 102 by embracing novel and singular categorisations of abuse into the remit of an exclusionary or exploitative category of consumer data abuse.

Similar efforts to regulate the US digital markets have led to a fierce debate over the suitability of such regulatory interventions. The prevailing view is that there has been a clear enforcement gap[9] in the digital markets,[10] especially in conglomerate mergers, due to an 'exaggerated' concern[11] about enforcement errors of early intervention,[12] which calls for a return to the anti-trust law. A prominent advocate[13] for consumer justice, namely for privacy, autonomy, and well-being, has persuasively argued in favour of regulating the collection and surveillance of consumer data as an unfair competition method, which is the orthodox approach adopted by the DMA for unfair competition in the digital markets. Furthermore, the latest move of the US Department of Justice,[14] announcing its intervention against Google for monopolising digital advertising technologies, signals a welcome return to the law. However, the critics[15] of 'progressive' intervention dismissed radical anti-trust reforms without rigorous support in empirical studies, which could demonstrate unambiguous consumer harm, instead incorporating socio-political goals for anti-trust analysis. Unfortunately, such regressive views

[8] For a comprehensive analysis of the DMA, see Anca D Chirita, 'Data-Driven Unfair Competition in Digital Markets' (2022) 29 Boston UJ of Sci & Tech L, available at <Data-Driven Unfair Competition in Digital Markets by Anca D. Chirita :: SSRN>.

[9] See Shili Shao, 'Antitrust in the Consumer Platform Economy: How Apple Has Abused Its Mobile Platform Dominance' (2021) 36 Berkeley Tech LJ 412 on the antitrust enforcement gap since the 1970s; cf on a lack of empirical evidence of slow anti-trust enforcement, see William H Rooney and Timothy G Fleming, 'Time for a New Sherman Act? The Debate on Antitrust Reform in Historical Perspective' (2022) Colum Bus LR 32.

[10] cf on 'precautionary' EU enforcement approach to digital markets, Douglas W Arner, Giuliano G Castellano and Eriks K Selga, 'The Transnational Data Governance Problem' (2022) 37 Berkeley Tech LJ 649.

[11] Notably, Fiona Scott Morton, Kartikeya Kandula, and Karissa Kung, 'Do We Need a New Sherman Act?' (2022) 42 Colom Bus LR 50; according to anecdotal evidence, except for eleven prohibitions before 1974, none occurred thereafter, see OECD, *Roundtable on Conglomerate Effects on Mergers* (Background Note by the Secretariat, 2020) 18.

[12] While the US *laissez-faire* approach was too permissive, the EU approach to digital monopolies was too strict and innovation friendly, see Richard J Gilbert, *Innovation Matters: Competition Policy for the High-Technology Economy* (MIT Press 2020) 217.

[13] See Maurice E Stucke, 'Addressing Personal Data Collection as Unfair Methods of Competition' (2022) 439 U of Tennessee Legal Studies Research Paper Series, forthcoming in (2023) Berkeley Tech LJ.

[14] See US DoJ, *Justice Department Sues Google for Monopolizing Digital Advertising Technologies Through Serial Acquisitions and Anticompetitive Auction Manipulation, Google Subverted Competition in Internet Advertising Technologies* (Press release 23-84/2023), available at <Justice Department Sues Google for Monopolizing Digital Advertising Technologies | OPA | Department of Justice>.

[15] cf on anti-trust 'populism', see Joshua D Wright and Jennifer Cascone Fauver, 'Antitrust Reform and the Nirvana Fallacy: The Case Against a New Sherman Act' (2022) 72 Colom Bus L Rev 93 and 113.

Electronic copy available at: https://ssrn.com/abstract=4424967

disregard consumer interest in digital economic privacy and the subtleties of algorithmic price discrimination and consumer data combinations.

Without undermining the real case for prescriptive anti-trust rules, applicable to digital markets, this chapter proves how the European Commission (EC)'s early anti-trust intervention in such markets has been a test case for anti-trust adaptation without a radical overhaul of the existing legal framework.

A major criticism of Article 102 has been its highly unpredictable nature as these singular categorisations stood in isolation. Therefore, this chapter will consider various singular categorisations of earlier consumer data abuse cases to identify the original stream of abusive conduct and decide on their inclusion into one of the three main categories of exclusionary, exploitative or hybrid consumer data abuses. The exploratory exercise of this chapter, hopefully, would eliminate unpredictability by offering clear guiding principles of expected business conduct vis-à-vis both businesses and consumers.

2. The Exclusionary Abuse of Consumer Data

Exclusionary data abuse[16] has featured, although less prominently, in earlier cases of digital dominance based on an unfair competitive advantage related to proprietary data. The latter belongs to an innovative technology misused to restrict consumer choice[17] with an unfair data-driven competitive advantage. An alternative to prescriptive rules on equality for businesses requires the exploration of how the principle of non-discrimination applies to digital markets cases. This chapter argues that applying this principle supports predictability against an expansive enforcement prioritisation and ensures future flexibility.

A. Non-Discrimination v Equality of Opportunity and Equal Access and Treatment of As-Efficient Competitors of Digital Monopolists

Earlier cases concerning an exclusionary consumer data abuse upheld the principle of non-discrimination, which preserved the opportunity equality for businesses and the equal treatment of as-efficient competitors of digital monopolists. A historical interpretation[18] of Article 102(c) reveals that its scope of application prioritised monetary transactions with

[16] On the aggregated sharing of consumer data, see Anca D Chirita, 'The Rise of Big Data and the Loss of Privacy' in Mor Bakhoum, Beatriz Conde Gallego, Mark-Oliver Mackenrodt, and Gintarė Surblytė-Namavičienė (eds) *Personal Data in Competition, Consumer Protection, and Intellectual Property Law: Towards a Holistic Approach?* (Springer 2018) 173.

[17] See also Kurz (n 1) 333.

[18] On the historical interpretation, see Anca D Chiriță, 'A Legal Historical Review of the EU Competition Rules' (2014) 63 ICLQ 304-5.

Electronic copy available at: https://ssrn.com/abstract=4424967

business partners, which could disadvantage them. A dynamic interpretation of the concept of consumer data abuse in the context of digital markets recognises the transactional value of consumer data for free digital services or products.

(i) An Unfair Consumer Data-Driven Competitive Advantage

Most historical cases of economic discrimination relate to an unfair competitive disadvantage, such as dissimilar pricing or other trading conditions, including standard business terms. Due to a novel concept of consumer data-driven competitive advantage, a dynamic interpretation of digital discrimination has evolved to include an exclusionary effect on as-efficient competitors through unequal access to valuable consumer data, which belongs to digital monopolists. As this dynamic interpretation gained irreversible traction, under the exclusionary consumer data abuse, there is a breach of the above principle of opportunity equality for businesses.

A historical reading of the *Microsoft I* case[19] demonstrates that for as-efficient competitors, securing privileged access to valuable proprietary data held by a digital monopolist amounted to an unfair, artificial interoperable competitive advantage. In contrast, a dynamic reading of the *Microsoft I* case consolidates the view that such access was in the public interest, having successfully passed the scrutiny of the legal test of an exceptional circumstance in which such access to confidential proprietary data must be indispensable for the software developers[20] to offer alternative and compatible software. Microsoft refused indispensable access to interface information.[21] Furthermore, in the *Google Android* case,[22] a multiple bundling of 12 to 30 Google Search-Play Store apps[23] offered a competitive consumer-data advantage,[24] which restricted competition on the ground of 'the detriment of consumers'.[25]

In comparison, consumer data consolidation through mergers and acquisitions, such as the *Facebook/Giphy* merger, comes with a greater awareness that higher quality consumer

[19] Case T-2011/04 *Microsoft* EU:T:2007:289; EC, COMP/C-3/37792 *Microsoft* [2004].
[20] *Microsoft*, para 85: 'to ensure Microsoft's competitors can develop … products which interoperate with the Windows domain architecture … and viably compete with Microsoft's work group server operating system'.
[21] ibid para 83: 'Microsoft did not allow access to, or use of, the interoperability information covered by the No Patent agreement on 'reasonable terms' and subject to charging remuneration rates', see para 120.
[22] Case T-604/18 *Google & Alphabet v EC* (*Google Android)* [2022] ECLI:EU:T:2022:541, para 305.
[23] ibid para 524.
[24] ibid para 543.
[25] ibid para 570.

Electronic copy available at: https://ssrn.com/abstract=4424967

data commands a 'substantial'[26] and an unfair, competitive advantage for Facebook and Google. Similarly, in the *Facebook/Kustomer* case,[27] Facebook enjoyed a 'significant' competitive advantage due to its 'superior' access to consumer data being indispensable for targeted advertising purposes. In the *Google/Fitbit* case, enjoying 'privileged'[28] access to 'extensive' aggregated health and fitness consumer data was attractive for corporate exploitation.

In contrast, in the *Apple/Shazam* case,[29] the EC did not regard Shazam's incremental consumer data as a 'significant' competitive advantage to Apple, especially for the marginalisation of as-efficient digital music streaming apps. Similarly, in the *Google/Waze* case,[30] although Waze enjoyed a 'first mover' competitive advantage in mapping crowd-sourced[31] traffic data as opposed to Google Maps' satellite and street view consumer data, the new entity competed against Apple Maps, TomTom, and NavFree.

An exclusionary foreclosure effect on as-efficient competitors of Facebook using consumer data predictive analytics,[32] including market trends, questioned whether Giphy consumer data increment could amount to a competitive advantage.[33] Paradoxically, competitors should have offered greater consumer privacy. The *Google/Looker* case[34] missed the opportunity to recognise a significant; thus, 'important',[35] competitive advantage of Looker's artificial intelligence (AI) software offering an automated analysis and visualisation of consumer data analytics.

(ii) FRAND Terms of Access to Consumer Data

Consumer data is instrumental in gaining unfair competitive advantage to which as-efficient competitors aim to secure fair and equal access since such data is indispensable for business growth and success and fostering effective residual competition. Paradoxically, while fair,

[26] CMA, *Completed Acquisition by Facebook Inc (now Meta Platforms Inc) of Giphy Inc* (Final Report on the case remitted to the CMA by the CAT, 2022), *Facebook/Giphy*, para 5201 (d).
[27] CMA, *Anticipated acquisition by Facebook Inc of Kustomer Inc* (27 September 2021), para 140.
[28] EC, DG COMP, Case M9660 *Google/Fitbit* (17 December 2020), paras 912 and 399 and [94].
[29] EC, DG COMP, Case M8788 *Apple/Shazam* (6 September 2018).
[30] See formerly Office of Fair Trading, *Completed acquisition by Motorola Mobility Holding (Google Inc) of Waze Mobile Limited* (17 December 2013), *Google/Waze*, paras 37, 47, 55, 60 and 74.
[31] See Elena Argentesi, Paolo Buccirossi, Emilio Calvano, Tomaso Duso, Alessia Marrazzo, and Salvatore Nava, *Ex-Post Assessment of Meger Control Decisions in Digital Markets* (Lear Report for the UK CMA, 2019) 77.
[32] *Facebook/Giphy* (n 26) para 8108.
[33] ibid para 8108 (b).
[34] ibid paras 12 and 14.
[35] CMA, *Completed acquisition of Google LLC of Looker Data Sciences Inc: Decision on relevant merger situation and substantial lessening of competition*, *Google/Data Looker* (16 March 2020), paras 160 and 202.

Electronic copy available at: https://ssrn.com/abstract=4424967

reasonable and non-discriminatory (FRAND) terms of access address the exclusionary theory of harm to residual competition and as-efficient competitors, such access comes at the expense of exploitative harm to the consumers. Although in the *Google/Fitbit* case,[36] the EC recognised that the parties shared access to sensitive health and fitness consumer data with 'third parties', due to the vagueness of such unfair terms, consumers are unaware of the potential consequences of their bargaining power in the health insurance markets. In contrast, the UK Competition and Markets Authority (CMA) cleared Microsoft's acquisition of Nuance,[37] which is active in the supply of healthcare transcription software. The new entity can exploit[38] healthcare-specific speech data and access Microsoft's speech recognition engine. In the *Google Android* case,[39] the General Court, too, recognised access to consumer data and behaviour as key parameters of digital competition and its 'ecosystem'.

(iii) Algorithmic Discrimination

Similar to the FRAND terms, the *Google I (Shopping)* case[40] resembled an essential, universal search engine, digital infrastructure that was indispensable for as-efficient comparison shopping websites to compete effectively against Google. The aggressive promotion of Google Shopping occurred concomitantly with the exclusionary algorithmic differentiation of as-efficient competitors demoted to an unequal, biased, and inferior business ranking position. This discouraged the average consumer from looking further down the first search engine page and on subsequent pages.

However, algorithmic discrimination lacked monetary consideration. While, in theory, an unbeknown consumer data value stood for the abstract consideration for the measurement of substantive unfairness derived from zero-pricing, a more pragmatic approach to algorithmic discrimination must rely on procedural unfairness of standard conditions of access to indispensable interoperability or algorithmic data, respectively. The EC did not analyse or quantify the economic value of algorithmic consumer data. Could it be possible that the real monetary conversion value succeeds the corporate profits analysis or

[36] *Google/Fitbit* (n 28) para 822.

[37] CMA, *Acquisition by Microsoft Corporation of Nuance Communications Inc: Decision on relevant merger situation and substantial lessening of competition*, *Microsoft/Nuance* (31 March 2022), para 55.

[38] ibid paras 157-58.

[39] *Google Android* (n 22) para 115-16.

[40] EC, DG COMP, Case AT39740 *Google Search (Shopping)* (27 June 2017); Case T-612/17 *Google Shopping* (10 November 2021) ECLI:EU:T:2021:763, para 183; Case C-48/22 P pending appeal.

Electronic copy available at: https://ssrn.com/abstract=4424967

revenue streams extracted from the actual consumer data manipulation for sharing or selling? A recent answer based on current practices will be discussed later.

As several categorisations of 'self' preferencing, favouring or promotion[41] have not been interpreted in conjunction with the exclusionary algorithmic discrimination of as-efficient competitors to encapsulate a 'combination' of exclusionary anti-competitive conduct, such categorisations of abuse; therefore, ignored the principal value of equal treatment and opportunity for as-efficient competitors of digital monopolists.[42] The *Google I (Shopping)* case emphasised that equal treatment of as-efficient competitors exposed to algorithmic rankings becomes a business expectation, which the digital monopolies, with the the DMA as gatekeepers, can no longer ignore. Since a digital monopolist could engage in algorithmic discrimination, this novel form of digital abuse has risen via algorithmic technology.

Article 6(5) of the DMA[43] is one of the first examples of a regulatory prescriptive obligation to not offer 'favourable treatment' in ranking, indexing, and crawling and apply transparent, fair, and non-discriminatory conditions to such rankings. As the above-mentioned obligations looked to codify earlier cases for the preservation of legal certainty, the *Google I (Shopping)* case could apply to an expanding digital ecosystem where universal, commercial, or mobile search engines could represent valuable points of reference for future anti-trust cases.

Although a slightly contrasting scenario, the *Microsoft I* case highlighted how as-efficient software developers could miss the business opportunity to offer interoperable solutions with the Windows package. In contrast, Article 6(4) of the DMA[44] prescriptively mandates the gatekeepers to allow technical installation and effective use of third-party software applications using, or interoperating with, the Microsoft operating system or access to its software applications. Despite the scarcity of relevant interoperability case law, it is widely applicable within the digital industry and rapidly expanding to include app stores.

[41] cf Rooney (n 9) 41 against 'equal' business rights for as-efficient competitors leaving monopolists unable to prioritise their own offerings.
[42] Dismissive of self-favouring for exclusionary demotion while advancing a special responsibility to business equality and consumer fairness, see Anca Chirita, 'Abuse of Global Platform Dominance or Competition on the Merits?' (2021) 33 Loy Consumer LR 13.
[43] DMA [2022] OJ L 265/35.
[44] ibid.

Electronic copy available at: https://ssrn.com/abstract=4424967

An interim finding regarding the exclusionary consumer data abuse is that the historical context of Article 102(c), which primarily focuses on monetary transactions, maintains principled relevance by reference to a differentiated standard and a competitive disadvantage. However, a more substantive normative clarification of algorithmic discrimination in rankings and interoperability cannot survive without adding to the law. Therefore, a pragmatic solution could apply *a minore ad maius* Article 102's general abuse prohibition in conjunction with the prescriptive obligations imposed by the DMA to fill in the existing legislative gaps in the digital context of consumer data abuse. In case of continued dissatisfaction with the above solution, more principled abstract drafting could account for digital ecosystems and their future technological development.

B. Fundamental Business Freedoms of Action and Choice

Digital case developments, such as *Microsoft I (Windows Media Player)*,[45] *Microsoft II (Internet Explorer)*,[46] and *Google Android*,[47] consistently applied the freedom of choice to behavioural consumer bias, following a pre-installed package of complimentary software applications, which were once part of an overall package of a primary product, namely Microsoft Windows and Google Android operating systems, respectively. For example, Google Play Store tied the Search app and Chrome browser.[48] However, pre-installation made it difficult for as-efficient competitors[49] to attract queries, revenues, and consumer data needed to improve their services. Due to this 'status quo' user bias,[50] pre-installation represented an 'important' distribution channel for those users who confine themselves to pre-installed or default app choices, which cannot be removed. Thus, users did not download alternative apps.[51]

While consumers needed the Media Player, Internet Explorer, Google Chrome browser, and the Search engine, it is clear that alternative developers had no real prospect of entering a zero-priced digital market as they could no longer offer a higher quality option at a lower price. The digital zero-pricing paradox is a stark reminder that openness in the public interest has constantly been an enforcement priority of the abuse prohibition to preserve business

[45] *Microsoft v Commission* (n 19) para 961.
[46] EC, DG COMP, Case AT39530 *Microsoft – Tying* (6 March 2013).
[47] EC, DG COMP, Case AT40099 *Google Android* (18 July 2018); confirmed on appeal, see Case T-604/18 (n 22).
[48] *Google Android* (n 22) para 300.
[49] ibid para 294.
[50] ibid para 305.
[51] Ibid paras 546-47 and 567.

Electronic copy available at: https://ssrn.com/abstract=4424967

freedom of action to compete against digital monopolies. However, in the absence of any real residual competition, consumers pay for the entire package; thus, having no real prospect of lowering higher prices in a primarily zero-priced market subsidised by profitable revenues from secondary and tertiary markets.

Under Article 6(3) of the DMA,[52] for the preservation of both business freedom of action and consumer choice, it must be possible to uninstall complimentary software apps, except when such apps are essential for technical integration. For example, while it would be demanding to require consumers to assemble hardware, the latest regulatory developments empower consumers to confirm their choice for the installation of any alternative software apps.

However, such consumer empowerment comes at an added price. Notably, Microsoft has further segmented the pricing for the Windows operating system to differentiate the Office suite price. The latter requires an annual license renewal with Microsoft as the sole beneficiary since the life span of the hardware is often shorter than the Office suite. In addition, Office introduced innovative features, which have made the subscription fee expensive for the actual consumer experience. For example, Word's reading-aloud function has only three voice accents and its dictation function moves at a slow speed when assessed to work as an AI digital writer. Nevertheless, the price of such failed innovations burdens the consumers. On the positive side, Microsoft has implemented differentiated product pricing. Due to interoperability, consumers can choose to not instal Word and switch to free alternatives. Thus, Office switching must happen for better value for money. Currently, consumers pay Microsoft an extremely high price for the use of a blank digital sheet of paper.

Under Article 6(4) of the DMA,[53] technical installation and effective use of alternative software applications, as outlined above, must enable such alternatives to become the default choice, if chosen by the consumers. The same applies to software application stores. The above provision ensures that alternative software apps continue to be of interest to business developers while also safeguarding consumer choice.

While the regulatory approach of the DMA has successfully captured prior case experience and technological mediums, the classic prohibition of dominance abuse has dealt with such cases. Under Article 102(d), a reference to 'supplementary obligations' are

[52] DMA [2022] OJ L 265/35.
[53] ibid.

Electronic copy available at: https://ssrn.com/abstract=4424967

purposively applied to alternative software functionality, which only had a distant connection with the main Microsoft Windows operating system product. A literal interpretation of the above provision can apply to cases of contractual tying; however, a contextual interpretation of the technological development of design choice and interoperability would demand a more inclusive interpretation of technological tying. In the *Microsoft I* case, the interpretation of technological tying was successful without adding too much to Article 102(d) in conjunction with a relevant secondary provision included in Directive 91/250/EEC.[54] Nevertheless, there was a misunderstanding regarding the material form of abuse, namely instead of being a classical refusal to supply,[55] there was a duty of disclosure concerning the proprietary interface information to ensure software interoperability. In the *Lietuvos* case,[56] the Court of Justice distinguished between the destruction of an essential facility in the form of a railway track removal[57] and the classic refusal to grant access to an indispensable infrastructure, which is difficult to duplicate. One must sensibly reconcile undistorted competition with conflicting interests, such as the contractual freedom to deal with trading partners of a dominant undertaking for granting access to essential infrastructures and property rights, including including investments,[58] made by the dominant undertaking. As Advocate General, Rantos, noted, the risk of 'free riding'[59] appears when competitors of a dominant undertaking take advantage of such access rather than being incentivised for investing in infrastructure. Therefore, the Advocate General persuasively argued in favour of safeguarding undistorted competition 'in the interests of consumers'[60] and for the duty to supply where a refusal could cause considerable damage to competition and the 'interests of consumers'.[61]

The outcome of the *Microsoft I* case tipped the balance in favour of freedoms of business action and consumer choice as safeguarding an effective competitive process. The

[54] See Council Directive 91/250/EEC on the legal protection of computer programs [1991] OJ L122, replaced by Directive 2009/24/EC on the legal protection of computer programs [2009] OJ L111/16, recital 17's reference to 'without prejudice' to the application of the prohibition of abuse where a dominant undertaking refuses to make available information that is necessary for interoperability.

[55] For this distinction, see Jones (n 4) 510; Lianos (n 4) 991 on a refusal to provide interoperability with an order to disclose proprietary information for interoperability purposes.

[56] Case C-42/21 P *Lietuvos geležinkeliai v EC* (12 January 2023) ECLI:EU:C:2023:12, para 79.

[57] ibid paras 81 and 91 with emphasis on removal being an independent form of abuse.

[58] ibid para 86.

[59] Case C-42/21 P Lietuvos *geležinkeliai v EC*, Opinion of Advocate General Rantos (7 July 2022) ECLI:EU:C:2022:537, para 65.

[60] ibid.

[61] On the marginal role of 'consumer interest' in exploitative abuses under Article 102, see Maria Ioannidou, *Consumer Involvement in Private EU Competition Law Enforcement* (Oxford University Press 2015) 35.

Electronic copy available at: https://ssrn.com/abstract=4424967

paradox lies in having no palpable evidence of consumer exploitation, especially due to the zero-priced complimentary service, which was a pro-competitive benefit disguised in an artificially high-priced Windows package. However, following the anti-trust intervention against technological tying, one can witness that consumers continue to pay higher prices for Microsoft software, such as the Office Suite and One Drive cloud service. The price of consumer 'experience' data for using such software continues to lack transparency; therefore, the next anti-trust venture must identify a price tag for valuable consumer data.

The outcome of the *Google Android* case served the developers of alternative browsers and search engines to develop competitive offerings and enhanced consumer choice. Pending an anti-trust investigation, the *Apple* tying case[62] against pre-installed apps on iPhones would assist in greater consumer choice. Against the unfair competition, Article 6(6) of the DMA[63] has protected an obligation to not impose technical restrictions to switching between software apps or to alternative subscriptions. Although the inclusion of switching marks an evolutionary statement in the digital case developments, a retrospective analysis of Article 102(d) reveals that the historical provision was long overdue for its continued viability in the context of digital markets. An interpretation *a minore ad maius* of the above DMA provisions in conjunction with Article 102(d) could alleviate further issues surrounding legal certainty. Concomitantly, a dynamic interpretation of the earlier cases of digital tying in the light of business and consumer freedoms of action and choice, respectively, preserves predictability.

3. Exploitative Abuse of Consumer Data

Facebook[64] and *Google Android*[65] are the latest anti-trust cases dealing with an exploitative abuse of consumer data with pending investigations.[66] Previously, the *Google Shopping* case recognised that consumer data were monetary consideration and the *Google Android* case[67] acknowledged the combination of consumer data gleaned from multiple applications, such as Chrome for search data, Maps for location data, YouTube for experience data and Gmail for location data. The General Court confirmed that Android devices gathered valuable consumer

[62] See Shao (n 9) 378; Friso Bostoen and Daniel Mândrescu, 'Assessing Abuse of Dominance in the Platform Economy' (2020) 16 Eur Comp J 474.
[63] DMA [2022] OJ L 265/35.
[64] HRC Düsseldorf, Case VI-Kart 1/19 (V) *Facebook v BKartA*; Chirita (n 8) 9.
[65] *Google Android* (n 22).
[66] See *CMA investigates Facebook's use of ad data* (Press Release, 4 June 2021) on Facebook's collection and use of advertising data.
[67] *Google Android* (n 47) paras 109, 110, 114, 739 and 997.

Electronic copy available at: https://ssrn.com/abstract=4424967

data, such as location or Google Play usage data[68] extracted from advertising. Similarly, in the *Privacy Sandbox* case,[69] the CMA demonstrated greater awareness of consumer privacy when accepting several commitments from Google, such as non-disclosure of consumer data gleaned from Chrome browsing history, analytics, third-party cookies for targeted advertisements, software codes, algorithms and business secrets, non-tracking and non-discrimination,[70] including self-preferencing.

In the age of consolidation of digital monopolies, the exploitative abuse of consumer data combinations[71] should not take anyone by surprise. In 1890, the US Sherman Act included the culprit of combinations of 'capital' organised in trusts,[72] which interchangeably, mirrored consumer data, as the latest online exchange currency for goods and services. However, following the EEC's enactment of the Merger Control Regulation in 1989, numerous digital mergers and acquisitions went under the radar in the form of pro-competitive conglomerates. In the US, Google's acquisition of Nest Labs[73] is an example of the importance of buying relevant consumer data combinations gleaned from Nest smart speakers and displays, thermostats, smoke and CO detectors, routers, and security solutions, such as surveillance cameras and doorbells. Since large digital conglomerates are insidiously behind an increased accumulation of wealth[74] and inequality,[75] it is right for the US Federal Trade Commission to review the GAFAM (Google, Apple, Facebook, Amazon, and Microsoft) past acquisitions.[76]

[68] *Google Android* (n 22) para 610. The Court dismissed the idea that Google had to recover its investment in Android at para 630.
[69] CMA, Case 50972 *Google Privacy Sandbox* Commitments (4 February 2022) [9] and paras 25-7.
[70] ibid para 30.
[71] For an earlier warning against the risk of data combinations for targeted advertisements, see Joseph Turow, 'Cracking the Consumer Code: Advertisers, Anxiety, and Surveillance in the Digital Age' in Kevin D Haggerty and Richard V Ericson (eds) *The New Politics of Surveillance and Visibility* (University of Toronto Press 2006) 299.
[72] See William L Letwin, 'Congress and the Sherman Antitrust Law: 1887-1890' (1956) 23 U of Chicago LR 248.
[73] See *Google Inc/Nest Labs* decision 20140457/2014 unavailable at <20140457: Google Inc.; Nest Labs, Inc. | Federal Trade Commission (ftc.gov)>.
[74] According to empirical evidence, between 1985 and 2019 real monopoly wealth increased by $25.36 trillion, see Kurz (n 1) 243.
[75] Sceptical of the nexus between economic concentration and wealth inequality, see Ioannis Lianos, 'Competition Law for the Digital Era: A Complex Systems' Perspective' (2019) 6 UCL Centre for Law, Economics and Society Research Paper Series 25.
[76] US, *FTC to Examine Past Acquisitions by Large Technology Companies: Agency Issues 6 (b) Orders to Alphabet Inc, Amazon.com Inc, Apple Inc, Facebook Inc, Google Inc, and Microsoft Corp* (Press release, 11 February 2020).

Electronic copy available at: https://ssrn.com/abstract=4424967

For the digital conglomerates, the counterfactual scenarios and the traditional theories of harm[77] demonstrated the culprit of consumer 'data combinations';[78] however, they did not eliminate the problem because the competition authorities did not opt for a narrower market definition,[79] namely a separate market for analytics,[80] sharing or selling of consumer data. For instance, the acquisition of Looker[81] offered Google an opportunity to access AI software to analyse aggregated consumer data for 'internal business use'.[82] The *Google/Looker* merger advanced several combinations of valuable consumer data generated by Google advertising, marketing, and cloud storage,[83] jointly with Looker warehouse and analytics services, to compete against Microsoft, Amazon, Teradata and Oracle[84] and facilitate the cross-selling[85] of complementary products.

A market for consumer data[86] could be instrumental in finding the reason for the rise of digital conglomerates. Once mega conglomerates easily received the green light; that is, without meaningful conditions or helpful remedies,[87] a clear conflict of interest appeared between harmful data mergers and the consumer data abuse. A more desirable outcome would have been the prohibition[88] of digital conglomerates, which are detrimental to consumer welfare, even in the advent of zero-priced services, because of the anti-competitive effects of consumer data combinations. A persuasive argument against structural remedies is that even successors may not be interested in protecting valuable consumer data.[89] While addressing consumer data abuse takes years of anti-trust investigations, preventing data combinations in the first instance is the ultimate solution. An anti-trust codification of harmful

[77] Superbly arguing for a conglomerate theory of consumer harm based on consumer data combinations, see Massimo Motta and Martin Peitz, 'Big tech mergers' (2021) 54 Information Econ and Policy 12.
[78] See 'Lear' report (n 31) paras I142 and I144-46.
[79] Anca D Chirita, 'Data-driven Mergers under EU Competition Law' in Orkun Akseli and John Linarelli (eds) *The Future of Commercial Law: Ways Forward for Change and Reform* (Hart 2020) 154.
[80] Except when *Telefónica Hutchinson* identified a market for mobile data analytics due to a larger customer base in the retail market; on appeal, see *CK Telecoms UK Investments Ltd v EC* (n 2) para 422; pending on appeal, Case C-376/20 P.
[81] *Google/Looker* (n 35) paras 12 and 14.
[82] ibid para 23 by reference to 'data warehouse solutions'.
[83] ibid paras 26, 45 and 120.
[84] ibid para 29.
[85] ibid para 28.
[86] Questioning the value of consumer data within various business models, see Pierre Regibeau and Ioannis Lianos, 'Digital Mergers: A Primer' (2021) 3 UCL Centre for Law, Economics and Society Research Paper Series 65.
[87] In favour of drastic break-ups of digital monopolies, see Kurz (n 1) 338.
[88] Brilliantly endorsing the prohibition of digital conglomerates, see Motta (n 77) 8.
[89] See Gilbert (n 12) 242.

Electronic copy available at: https://ssrn.com/abstract=4424967

data combinations can securely transpose consumer data combinations into policy prescriptions.

The *Google/Double Click* case[90] represented a missed opportunity to identify the hidden motive of the 'killer' acquisition through which Google effectively monopolised the publisher ad server (double click for publishers) and the exchange technology (double click ad exchange, known as AdX) for auctions to increase advertising fees by 20% and lower the choices.[91] Thereafter, Google implemented several secret projects. First, under Project Bernanke,[92] Google manipulated AdX auction bids to consolidate its digital dominance towards the detriment of as-efficient competitors and tweaked its algorithm against publishers who used innovative header-bidding technology.[93] By preventing advertisers from purchasing alternative inventory, Google eliminated as-efficient competitors and made publishers dependent on Google ad server and exchange technologies.[94] Furthermore, Google increased its competitive 'data' advantage by acquiring privileged access to rival bids[95] and used it to adjust its fees and extract higher corporate profits.[96] Second, under Project Poirot,[97] Google won rival header-bidding bids sent to as-efficient ad exchanges by artificially manipulating them below the minimum price agreed upon by the advertisers.[98] Third, under Google's 'Unified Pricing Rules',[99] publishers could not set different prices for ad exchanges or advertising, which could lessen Google's ad server quality.[100] Fourth, under Project Bell, Google artificially lowered rival publisher bids.[101] Therefore, by implementing the above strategies for over two decades,[102] Google imposed[103] its business terms on publishers and advertisers towards the detriment of competition and consumers.[104] Similarly, the EC

[90] EC, DG COMP, Case M4731 *Google/DoubleClick* [2008].
[91] US District Court for the Eastern District of Virginia, Alexandria Division, Case 1:23-cv-00108 (24 January 2023) under §4 of the Sherman Act, §15 USC para 304; also, paras 80, 22, 16, 20, 266, 274 and 312 (1), respectively.
[92] ibid paras 24, 75 and 128 on appropriating publishers' revenues.
[93] ibid para 161.
[94] ibid paras 78, 107 and 91, respectively.
[95] ibid para 184.
[96] ibid para 198.
[97] ibid paras 216, 217, 218 and 312 (9).
[98] ibid paras 221, 262 (9) and 274.
[99] ibid paras 239 and 312 (10).
[100] ibid para 245.
[101] ibid paras 263 (7) and 312 (7).
[102] ibid para 264.
[103] ibid para 262.
[104] ibid para 315.

Electronic copy available at: https://ssrn.com/abstract=4424967

investigates whether Facebook artificially tied its Marketplace ad service[105] to gain an unfair competitive advantage, exclude competitors and impose unfair trading conditions on rival advertisers on both Facebook and Instagram.

The emerging exploitative abuse of such conglomerate consumer data combinations falls under Article 102 or, as mentioned above, under the US Sherman Act's monopolisation claims. As there is no specific provision on the consumer data combination, this is a digital dominance gap. Furthermore, consumer data combinations are a consequential form of abuse of dominance based on the merit of valuable consumer data, worsened by inadequate handling of digital mergers involving a large accumulation of consumer data. In the *Google/Fitbit* case,[106] the EC considered that aggregated consumer data combinations complied with data protection laws. However, due to large amounts of sensitive consumer data combinations, improved commitments[107] had to alleviate the exclusionary foreclosure in emerging healthcare markets.

In the *Facebook* case,[108] Article 102(a) on unfair conditions was instrumental in highlighting the consumer's weaker bargaining power. By empowering consumers, their freedom of choice could prioritise economic privacy over personalised experience, especially for those who do not value personalised services. Similarly, in the *Epic Games* case,[109] a consumer class action was introduced under Chapter II prohibition in §18 of the UK Competition Act of 1998 against Apple for its exploitative abuse of restrictive standard business terms and conditions imposed on app developers, unfair prices[110] for its distribution of apps via the App Store, in-app purchase functionality and removal of the Fortnite app from the App Store. In addition, the EC investigates whether Apple[111] abused its dominant position by forcing music streaming app developers to accept its payment app and restricting them from recommending alternative music subscriptions. Without mentioning Article 102(a), the

[105] EC, *Antitrust: Commission sends Statement of Objections to Meta over abusive practices benefiting Facebook Marketplace* (Press release IP/22/7728, 19 December 2022).
[106] *Google/Fitbit* (n 28) para 411.
[107] ibid para 921.
[108] *Facebook v BKartA* (n 64); Chirita (n 8) 9.
[109] See Joined Cases 1377/5/7/20 and 1378/5/7/20, *Epic Games Inc, Epic Games International SARL and Epic Games UK Ltd v Apple Inc and Apple (UK) Ltd*; *Epic Games Inc and Epic Games International SARL v Alphabet Inc, Google LLC, Google Ireland Ltd, Google Commerce Ltd, Google Payment Ltd* (UK CAT, 22 February 2021, Roth LJ), paras 49 and 50 (b).
[110] ibid paras 70 and 120, suggesting a 30% commission fee imposed on developers.
[111] EC, *Antitrust: Commission sends Statement of Objections to Apple clarifying concerns over App Store rules for music streaming providers* (Press release IP/23/1217, 28 February 2023).

Electronic copy available at: https://ssrn.com/abstract=4424967

EU Commissioner for Competition[112] highlighted that Apple and Meta engaged in unfair trading practices, namely unfair and exploitative trading conditions imposed on as-efficient competing developers. Beyond allegations of exclusionary tying of pre-installed apps on Android devices, the CMA investigates Google's[113] and Apple's[114] app stores for imposing unfair conditions on as-efficient competing app developers, such as the acceptance of their payment apps.

In the *Meta* case,[115] the CJEU, too, has opened the door to consumer class actions for infringement of data protection laws based on national unfair competition or consumer law, including unfair terms and conditions. Cases of exploitative abuse are fraught with difficulties associated with the burden of proof. In the *Amazon* case,[116] a UK consumer class action for damages started against Amazon Buy Box's[117] unjustified advertisement as 'the cheapest' and prominently 'Featured Offer' in Amazon search listings, especially when tied to its privileged Prime delivery and logistics services, due to its biased 'Merchant Algorithm'.

In a landmark consumer class action against *Meta*, under §18 (2)(a) of the UK Competition Act of 1998,[118] Facebook's exploitative abuse of consumer data focused on its sharing of sensitive consumer data for personalised targeted advertisements. Unfortunately, the danger arising from the sale of aggregated consumer data was trivialised as a legitimate form of 'direct monetisation',[119] as opposed to the indirect share of consumer data. The legal standard for procedural unfairness of Facebook's business terms regarding the sharing of aggregated sensitive consumer data relied on the theory of consumer's weaker bargaining power; the latter must accept imposed terms on a 'take it or leave it' basis;[120] therefore, enjoying no real autonomy and freedom of choice to opt out of a personalised experience for

[112] EC, *Keynote delivered by EVP Vestager for the Keystone Conference: A Triple Shift for competition policy* (Speech/23/1342, 2 March 2023).
[113] CMA, *Investigation into suspected anti-competitive conduct by Google* (10 June 2022), last updated on 10 March 2023.
[114] CMA, *Investigation into Apple AppStore* (4 March 2021), last updated on 10 March 2023.
[115] See Case C-319/20, *Meta Platforms Ireland Limited v Bundesverband der Verbraucherzentralen and Verbraucherverbände – Verbraucherzentrale Bundesverband eV* (28 April 2022) ECLI:EU:C:2022:322, para 83.
[116] See Case no 1568/7/7/22 *Julie Hunter v Amazon.com Inc and Others*, 'Notice of an Application to Commence Collective Proceedings under §47B of the Competition Act 1998' (UK CAT, 4 April 2023).
[117] On Amazon's Buy Box algorithm, see Karishma Puri, 'Is Bezos' Patience his virtue or his vice? Applications of US antitrust and EU competition law to *Amazon.com*'s consumer-focused corporate strategy' (2022) 6 Durham LR 27 ff.
[118] See Case no 1433/7/7/22 *Meta Platforms Inc, Meta Platforms Ireland Ltd, and Facebook UK Ltd* (UK CAT, 20 February 2023, Sir Smith J), paras 11, 12 and 18.
[119] ibid para 13.
[120] ibid para 21.

Electronic copy available at: https://ssrn.com/abstract=4424967

purely subjective corporate considerations. There is further evidence for procedural unfairness of 'vague' standard terms about the destination of aggregated consumer data, especially where there is an unclear purpose for its use and potential for abuse. Therefore, Facebook users must know in advance that it shared consumer data, particularly, if it sold such data. Following the theory of informed consent of the average Facebook users who are not in a position to make sense of sophisticated legal terms that are smartly drafted to disadvantage them, there is no justification for 'excessively long and/or complex' terms.[121] This is an exploitative consumer data abuse due to Facebook's monopolistic digital power. However, by downgrading the above claims to unfair consumer contract terms tainted with 'misleading'[122] representations, which are independent of the theory of monopolistic abuse of digital dominance, this interpretation is incongruent with the Consumer Rights Act of 2015.

As the proposed counterfactual scenario examined how such consumer data would not be 'extracted'[123] for an unlawful purpose, Sir Justice Marcus Smith emphasised the difficulty associated with the recovery of a gain from trade with aggregated consumer data, as opposed to sustaining an economic loss, which was the case here due to a lack of financial compensation[124] for valuable consumer data in lieu of monetary consideration.[125] If such economic loss were impossible to quantify due to an 'unfairly low',[126] that is, zero-priced service, it would still be possible to demonstrate how Facebook accrued gains from the sharing and trading of aggregated consumer data. Although this class action benefits selected consumers, disclosing corporate gains from the sharing of sensitive consumer data is in the public interest. Surprisingly, the UK Competition Appeal Tribunal (CAT) relied on the 'binding'[127] test, which was set up in *United Brands* for classical excessive pricing, rather than a flexible application of substantive unfairness to freemium digital services for which consumers paid with aggregated; thus, sensitive consumer data. The rationale for the CAT's choice was its futile efforts to elucidate the *United Brands* test in *CMA v Flynn* as this test lacks a clear methodological[128] framework for consumer loss analysis.

[121] ibid para 29.
[122] ibid para 30.
[123] ibid para 22 (3) (ii) and (iii).
[124] ibid para 23 at 128 (a)'s reference to a 'recompense'.
[125] ibid para 25 (1).
[126] ibid paras 23 and 24.
[127] ibid para 28.
[128] Ibid para 35.

Electronic copy available at: https://ssrn.com/abstract=4424967

While an outdated test was unsatisfactory for the excessive pharmaceutical pricing, there are dissimilar facts in the present case. Despite the well-known limitations of the *United Brands* test for excessive pricing, the interpretation of substantive unfairness in *Flynn*[129] relied on a lack of 'reasonable' relation to the 'economic' value of the service. Applying this interpretation to freemium digital services disadvantages consumers who are unable to prove economic loss. This demonstrates that the *United Brands* test[130] is unsuitable for the actual quantification of monetary value. Furthermore, it is this lack of monetary payment instead of service subscription, which can help prove a causal relationship[131] between the former and consumer data for targeted advertisements. The fact that paid subscriptions could only diminish Meta's monopolistic corporate profits,[132] raises eyebrows as the most untenable proposition for consumer justice. In addition, it would be cumbersome, *ad absurdum*, to prove how a price of zero is 'too high'.[133] This inquiry dismissed the case for consumer justice, which should not be about the excessiveness of zero-pricing (*probation diabolica* for substantive unfairness), but rather about an excessive disclosure of consumer data (procedural unfairness pragmatism).

Therefore, one cannot interpret the absence of Meta's direct imposition of a price tag as a lack of evidence of substantive unfairness. The prohibition of abuse of dominance captures the imposition of unfair pricing, which occurs either 'directly or indirectly'. The absence of an unequivocal test for substantive unfairness should not be a shield for monopolistic unfairness. At trial, judges must overcome the difficulties presented above to emphatically establish a UK precedent that recognises consumer loss of digital economic privacy, inclusive of consumer location and willingness to pay.[134]

4. Consumer Justice through a Hybrid Consumer Data Abuse – For Greater Autonomy, Enhanced Bargaining, and Consumer Empowerment

[129] See *CMA v Flynn Pharma Ltd* [2020] EWCA Civ 339, para 61.
[130] The same goes for the alternatives advanced in *Napp Pharmaceutical Holdings Limited v Director General of Fair Trading* [2002] CAT 1, para 390.
[131] cf Case no 1433/7/7/22, *Meta Platforms Inc, Meta Platforms Ireland Ltd, and Facebook UK Ltd* (UKCAT, 20 February 2023) para 47, page 36.
[132] ibid para 47 (iii), para 38.
[133] ibid para 52.
[134] See the excellent contribution of Noga Blickstein Shchory and Michal S Gal, 'Voice Shoppers: From Information Gaps to Choice Gaps in Consumer Markets' (2022) 88 Brooklyn LR 43, recognising algorithmic price discrimination based on consumer 'willingness to pay' for voice shoppers.

Electronic copy available at: https://ssrn.com/abstract=4424967

Anti-competitive foreclosure, leading to consumer harm, is an inspiration for hybrid consumer data abuse, especially for an analytical inclusion of both exclusionary and exploitative consumer data abuse. A dualistic abuse of consumer data must be built upon a narrower definition of consumer privacy.[135] As the prevailing digital currency, consumer privacy must be inclusive of consumer autonomy over economic preferences, interests, behaviour, and choices, and provide an enhanced consumer bargaining of reservation prices with further consumer empowerment for physical and virtual locations, which could predict their disposable income. This narrower interpretation of consumer privacy responds to earlier criticisms[136] that consumer anti-trust could destabilise the exclusivist focus on price competition. In other words, for greater economic privacy, consumer data flexibility must replace price rigidity.

As discussed earlier, consumer privacy was developed to counteract digital monopolies' offerings of free services, which are paid for with consumer data that were either shared or sold to advertisers and merchant businesses and retailers, respectively. Therefore, the economic context of targeted advertising highlighted in the *Meta* case[137] for multiple legal combinations of consumer data focused on consumer data shared for personalised sales. Personalisation or consumer data-driven algorithmic discrimination exploits consumer ignorance for unwanted attention, namely, advertisements. Due to such unprecedented personalisation, predictive consumer data analytics and behaviour have significantly improved.[138]

Examining the legal context of consumer data combinations in the *Facebook/Giphy* case,[139] the CMA placed significant value on ways of selling consumer data for higher prices, especially for personalised targeted advertisements, to the customers. In the *Facebook/Kustomer* case,[140] the CMA noted that the advertisers were interested in consumer characteristics, location, language, and culture. Similarly, in the *Google/Fitbit* case,[141] the EC

[135] See Chirita (n 8) 27.
[136] Maureen K Ohlhausen and Alexander Okuliar, 'Competition, Consumer Protection, and the Right (Approach) to Privacy' (2015) 80 Antitrust LJ 156.
[137] Case C-252/21 *Meta Platforms Inc v Bundeskartellamt and Verbraucherzentrale*, Opinion of AG Rantos (22 September 2022) ECLI:EU:C:2022:704, para 49.
[138] See Filippo Lancieri and Patricia Morita Sakowski, 'Competition in Digital Markets: A Review of Expert Reports' (2021) 26 Stanford J of Law, Bus & Finance 84, highlighting interoperability as essential to attract scale; thus, subject to limitations that preserve digital dominance.
[139] *Facebook/Giphy* (n 26) para 5201 (d).
[140] *Facebook/Kustomer* (n 27) para 119.
[141] *Google/Fitbit* (n 28) paras 448 and 455.

Electronic copy available at: https://ssrn.com/abstract=4424967

predicted that Fitbit consumer data about health, wellness, and location could significantly improve targeted advertisements by capturing Apple iPhone users who actively use Fitbit instead of the Google Fit health-tracking mobile app.

Unequivocally, the above sharing or selling of consumer data, for algorithmic price discrimination[142] based on consumer preferences, interests, choice, and location, could exploit consumer vigilance, especially for bargaining the reservation price.

A consumer welfare paradox arises when competition authorities, notably, the US Federal Trade Commission's recognition[143] of product design improvements, rely on consumer law to justify exclusionary conduct; however, rigidly endorses an exceedingly high burden of proof for exploitative pricing effects on consumer welfare. In the absence of a carefully weighted counterfactual analysis of consumer purchase price increases vis-à-vis the added margin caused by consumer data exploitation, it is unrealistic to suggest that personalised offers are free, excluding the marketing costs associated with personalised advertisements. Therefore, the above lack of evidence of a marginal price increase and unrealistic counterfactuals have consolidated aggregated consumer data combinations following strategic mergers and acquisitions. Inter alia, in the *Facebook/Instagram* case,[144] there was no lessening of competition because Instagram was unable to compete against Facebook, dismissing the potential for better-targeted advertising[145] due to a consumer data combination.

Leaving aside the existential concerns about the future incentives of digital monopolies to supplement rather than cut human labour,[146] Microsoft Outlook introduced automated AI-driven email replies. This novel feature proved how Microsoft actively engaged in email data mining. Outlook users are automatically signed-in to allow this feature, which is a breach of business confidentiality and user privacy. This demonstrates that personalised offers sacrifice business confidentiality and consumer privacy through automated AI techniques, which combine Outlook, Word, and OneDrive consumer data for corporate profits. Due to the multitude of Microsoft services, and following its privacy policy,[147] the large-scale

[142] For an extensive analysis of behavioural discrimination, see Chirita (n 8) 25.
[143] See MR Patterson, *Google, Yelp, LIBOR, and the Control of Information* (Harvard University Press 2017) 122.
[144] See OFT, *Anticipated acquisition by Facebook Inc of Instagram Inc* (ME/5525/12, 14 August 2012) para 29.
[145] ibid para 34.
[146] In the same vein, Kurz (note 1) 37.
[147] See Microsoft's Privacy Statement, last updated: February 2023, available at <Microsoft Privacy Statement – Microsoft privacy>; however, Outlook's privacy terms are opaque.

Electronic copy available at: https://ssrn.com/abstract=4424967

combination of consumer data is an asset for competitive advantage, which leverages Windows' dominance into emerging Office suite and cloud solutions. So far, Microsoft's anti-competitive conduct has been under the radar of competition authorities, with *Bundeskartellamt*'s latest investigation[148] being a notable exception.

Therefore, the exponential price increase must capture the vast trove of consumer data relevant to merchants, retailers, and other sellers. The traditional 'Small but Significant Non-Transitory Increase in Price' (the SSNIP) test[149] has often been, too easily, dismissed[150] in favour of a novel 'Small but Significant Non-Transitory Decline in Quality' (the SSNDQ) test[151] to capture the loss of consumer privacy as an attribute of quality. Commentators doubt the suitability of the SSNDQ test to deal with consumer data combinations.[152] Dismissing the classical hypothetical monopolist test in the *Google Android* case, the General Court endorsed the EC's application of the SSNDQ test[153] without requiring a 'precise standard of degradation'[154] to ascertain whether Apple exerted a competitive constraint on Google. Although consumers were not 'insensitive'[155] to a small deterioration in Android's quality, they were unlikely to switch[156] to a non-licensed operating system. App developers,[157] too, had an interest in targeting the most widely used platform, which undermined the prospect of an alternative and credible[158] version of Android. Similarly, in the *Facebook/Giphy* case,[159] the risk of negative publicity, caused by the misuse of consumer data and privacy abuse, stood for a good metric of lower quality in the absence of meaningful consumer switching from the dominant platform. Although consumers appear immune to privacy decline, this is due to the

[148] See Bundeskartellamt, *Bundeskartellamt/Digital economy: Investigation of Microsoft's significance for competition across markets* (Press release, 28 March 2023) concerning the integration of AI into Microsoft services, such as its Azure and One Drive cloud, Teams conferencing app, and xbox gaming, among others.
[149] Critical of SSNIP arbitrariness, see Viktoria H S E Robertson, *Competition Law's Innovation Factor: The Relevant Market in Dynamic Contexts in the EU and the US* (Hart Publishing 2020) 252.
[150] This is due to 'bandwagon' effects, see Lancieri (n 138) 80.
[151] For pertinent criticism of the earlier SSNDQ assumptions, see Thomas Stuart, 'Too Little Too Late? An Exploration and Analysis of the Inadequacies of Antitrust Law When Regulating GAFAM Data-Driven Mergers and the Potential Legal Remedies Available in the Age of Big Data' (2021) 17 Eur Comp J 413.
[152] See Jacques Crémer, Yves-Alexandre de Montjoye, and Heike Schweitzer, 'Competition Policy for the Digital Era: Report for the EC' (2019) 121; Jean Wei Lee (Kit), 'Data-Driven Mergers: Is It Time to Reform EU Merger Control?' (2023) 6 De Lege Ferenda (Cambridge LR) 118.
[153] *Google Android* (n 47) paras 141-42 and 177.
[154] *Google Android* (n 22) para 180.
[155] ibid paras 191 and 193.
[156] ibid para 219.
[157] *Google Android* (n 22) paras 218 and 224.
[158] ibid para 229.
[159] *Facebook/Giphy* (n 26) para 5118.

Electronic copy available at: https://ssrn.com/abstract=4424967

misunderstanding about how an economic privacy loss could lead to higher prices. Since the CMA predicted that the *Facebook/Giphy* merger[160] would lead to a substantial lessening of competition, it ordered Facebook to sell Giphy. Both the outcome and the remedy marked a significant departure from the earlier lenient treatment of digital conglomerates.

Nevertheless, if one were to assume that in a hypothetical worst-case scenario, most visible and affluent[161] consumers who overshare consumer data about their economic interests, preferences, and choices on monopolistic platforms must subsequently pay, on average, up to 15 to 25% higher prices due to having their digital footprint analysed for the prospect of exploitative price increase, then the SSNIP test could be helpful. It could examine the average price, which less visible and non-affluent consumers who go for traditional shopping rather than digital platforms, even for leisure purposes, pay for similar offerings. Extracting the valuation difference could be the key to the said price increase. Even a modest price increase,[162] within a range of 5 to 15%, does not offer a long-term lower price guarantee for consumers.

Therefore, in the *Amazon/Deliveroo* case,[163] the CMA was concerned about the effects of behavioural price discrimination on the supply of online food platforms and logistics as both parties shared consumer data. For its logistics,[164] Deliveroo relies on AI predictive analytics of historical and external consumer data to offer an estimated delivery time to consumers and analytics services to restaurants. In contrast, as-efficient competitors, such as Pizza Hut, lacked shared access to aggregated consumer data,[165] which is essential for

[160] See CMA, *Completed acquisition by Facebook Inc (now Meta Platforms Inc) of Giphy Inc* (Summary of Remittal Final Report, 2022) para 3.

[161] Another highly provocative fairness question is whether price discrimination could lead to higher prices for affluent consumers, as opposed to lower prices to less well-off consumers, see Joost Poort and Frederik Zuiderveen Borgesius, 'Personalised Pricing: The Demise of the Fixed Price?' in Uta Kohl and Jacob Eisler (eds) *Data-Driven Personalisation in Markets, Politics and Law* (Cambridge University Press 2021) 186.

[162] On the dissatisfaction of the majority of 'dependent'; thus, loyal, consumers who were imposed exploitative higher prices through behavioural price discrimination, see Jun Wang, Tao Shu, Wenjin Zhao and Jixian Zhou, 'Research on Chinese Consumers' Attitudes Analysis of Big-Data Driven Price Discrimination Based on Machine Learning' (2022) 12 Frontiers in Psychology 12; on much higher prices unfairly charged to loyal customers, see Lupin Sun and Yanfeil Tang, 'Data-Driven Discrimination, Perceived Fairness, and Consumer Trust – The Perspective of Consumer Attribution' (2021) 12 Frontiers in Psychology 2 and 10; thus, consumers who are aware of price discrimination are buying less, see Alexandre de Cornière and Rodrigo Montes, 'Consumer Privacy and the Incentives to Price-Discriminate in Online Markets' (2017) 16 R of Network Econ 294; due to online price discrimination, consumers are worse off, see Wen-Chung Guo and Fu-Chuan Lai, 'Price discrimination under online-offline competition' (2022) 2016 Economic Letters 1.

[163] CMA, *Anticipated acquisition by Amazon of a minority shareholding and certain rights in Deliveroo* (4 August 2020) para 928 (b) (ii).

[164] ibid paras 320, 322 and 324.

[165] ibid para 565.

Electronic copy available at: https://ssrn.com/abstract=4424967

restaurants.[166] For Amazon's consumer-centric global strategy,[167] the speed of its delivery service is of paramount importance and represents a key competition driver for consumers who prioritise convenience and indicate a greater willingness to pay.[168] The CMA accepted that the new entity could offer a faster delivery alternative for such consumers; thus, others may still prefer local shops. Although there was noticeable price competition between online and offline groceries,[169] Amazon Prime customers paid comparable supermarket prices with no delivery fee above a certain threshold. Therefore, under its Phase I investigation, the CMA cleared this transaction as not lessening competition.

In contrast, scholars argued that ethically, algorithmic price discrimination was fairer[170] than classical discrimination. Others welcomed an algorithmic discrimination regulation[171] to alleviate the current digital asymmetry where monopolists engage in exploitative abuses of extensive consumer data. Such permanent monopolies have combined economic power,[172] raised wealth inequality,[173] and eroded democracy. Yet, monopolistic extensions of digital conglomerates have, for long, followed a favourable theory of price discrimination where bundling and tying[174] do not have adverse effects on consumer welfare. This is a flawed position, which has led to enforcement discrepancies. For example, Microsoft received two fines for its anti-competitive digital tying cases with no similar prospect contemplated for its acquisitions of Skype, LinkedIn, or Nuance.[175] Similarly, Google acquired Looker despite interoperability concerns regarding rival data warehouses.[176]

Nonetheless, there is greater recognition that algorithmic discrimination exploits consumer vulnerability.[177] It does this, first, through differentiated pricing based on consumer

[166] ibid para 584.
[167] ibid para 6112.
[168] ibid para 5182.
[169] ibid para 5183.
[170] See Yochanan E Bigman, Desman Wilson, Mads N Arnestad, Adam Waytz, and Kurt Gray, 'Algorithmic Discrimination Causes Less Moral Outrage Than Human Discrimination' (2023) 152 J of Experimental Psychology 24.
[171] See Mateusz Grochowski, Agnieszka Jablonowska, Francesca Lagioia, and Giovanni Sartor, 'Algorithmic Price Discrimination and Consumer Protection: A Digital Arms Race?' (2022) Tech and Regulation.
[172] On financialisation of market power, see Ioannis Lianos and Andrew McLean, 'Competition Law, Big Tech and Financialisation: The Dark Side of the Moon' (2021) 5 UCL Centre for Law, Economics and Society Research Paper Series 10.
[173] See Kurz (note 1) 48 suggesting the imminence of plutocracy and oligarchy.
[174] See OECD (n 11) 18.
[175] See *Microsoft/Nuance* (n 37) para 146; Chirita (n 16) para 175-76.
[176] *Google/Looker* (n 35) para 232.
[177] For an original analysis of voice shoppers, see Gal (n 134) 19.

Electronic copy available at: https://ssrn.com/abstract=4424967

data analysis of personal 'sensitivities' and, second, through consumer manipulation for expensive purchases. The latter is a dynamic variety of algorithmic pricing akin to consumer-pressured sales.

Algorithmic discrimination has risen whereby privacy, human rights, consumer, and data protection[178] laws cannot deal with the rise of behavioural price discrimination.[179] This has been echoed in the *Google/Fitbit* case.[180] Data protection, privacy, and the protection of communications confidentiality cannot alleviate the risk associated with the monopoly control of aggregated consumer data, which could have exclusionary and exploitative effects on rivals and consumers, respectively. Due to the granular collection of sensitive medical and wellness consumer data, such as heart rate, sleep activity and oxygen saturation level, cookies, and geolocation data;[181] consumer data exploitation for personalised advertisements was a far greater danger for strengthening Google's dominance in search advertising. As economic experts[182] acknowledged, Google leveraging its expansion strategy into healthcare insurance markets relies on the analysis of Fitbit's sensitive consumer health data through Google Cloud and Project Nightingale. Following a successful monopolisation of healthcare insurance,[183] overall[184] consumers could only be worse off due to short-lived merger-specific efficiencies. Therefore, the harmful effects of conglomerates on consumer welfare seeking an end to innovative pharmaceutical projects have been recognised as 'killer acquisitions', raising concerns about similar spill-over effects in the digital markets.[185] In her opinion,[186] Advocate General Kokott referred to such acquisitions of innovative start-ups at an early development stage in the field of internet services, pharmaceuticals, or medical technology

[178] On the limits of data protection, see Michèle Fink, 'Hidden Personal Insights Entangled in the Algorithmic Model: The Limits of the GDPR in the Personalisation Context' in Kohl (n 161) 105.
[179] In the same vein, see Alan M Sears, 'The Limits of Online Price Discrimination in Europe' (2019) 21 Colum Sci & Tech LR 14 and 37 persuasively arguing that competition law, consumer protection and even unfair competition cannot deal with online price discrimination; thus, Geo-Blocking Regulation could be helpful; for earlier signs of online price discrimination as 'web lining', as opposed to racial or 'redlining' discrimination, see Oscar Gandy JR, 'Data Mining, Surveillance, and Discrimination in the Post-9/11 Environment' in Haggerty (n 71) 366.
[180] *Google/Fitbit* (n 28) para 412.
[181] ibid paras 416-19, 423, 445-46.
[182] See Zhijun Chen, Chongwoo Chloe, Jiajia Cong and Noriaki Matsushima, 'Data-Driven Mergers and personalization' (2022) 53 RAND J of Econ 4.
[183] ibid 6-7.
[184] Arriving at the same conclusion for digital conglomerates, see Motta (n 77) 11.
[185] Generally, CM Cunningham, F Ederer and S Ma, 'Killer Acquisitions' (2021) 129 J of Political Econ 18 and 40.
[186] Case C-449/21, *Towercast v Autorité de la concurrence, Ministère de l'Économie and Others*, Opinion of Advocate General Kokott (13 October 2022) ECLI:EU:C:2022:777, paras 41 and 48.

Electronic copy available at: https://ssrn.com/abstract=4424967

and distinguished between an *ex-ante* 'preventative' merger control and a 'weaker' *ex-post* control of monopolies. Therefore, following the approval of data-driven mergers, as has happened in the *Google/Fitbit* case and others, the danger of algorithmic price discrimination based on aggregated, sensitive, consumer data should no longer be underestimated for its impact on consumer welfare.

Unfortunately, the impact of online price discrimination on consumer welfare has often been insufficiently analysed by competition authorities due to a lack of empirical datasets,[187] which could persuade such authorities that discrimination based on consumer data analytics has become widespread. In addition, due to econometric tools, such as the SSNIP, being unable to capture discriminatory prices, payments increase in the digital markets. While this is difficult for zero-priced services, it should be possible to detect algorithmic price discrimination for online shopping and sales.

5. Conclusions

This chapter advanced a novel concept of consumer data abuse, alongside algorithmic price discrimination and consumer data combinations, which have hybrid negative welfare effects, including exclusionary and exploitative effects on fundamental freedoms of business action and consumer choice, respectively.

While identifying legislative lacunae concerning the use of AI software algorithms for greater predictive analytics of consumer behaviour owing to the availability of consumer experience, location, medical and financial data and marketing strategies, such as personalised offers, this chapter has advanced legal interpretations to close the gap in the prohibition of abuse of digital dominance when applied in conjunction with the DMA provisions relevant to consumer data combinations, digital tying, and algorithmic discrimination.

Highlighting how digital conglomerate mergers and killer acquisitions helped sustain an unfair consumer data-driven competitive advantage coupled with consumer data combinations towards the detriment of as-efficient competitors of digital monopolists and consumers, this chapter advanced the conflicts of interest between the under-enforcement of the consumer data abuse and the lenient scrutiny of data-driven mergers. In addition, it

[187] See also Lancieri (n 138) 99 and 168.

Electronic copy available at: https://ssrn.com/abstract=4424967

highlights the conflicts between FRAND terms of access to aggregated, sensitive consumer data and consumer economic privacy.

Furthermore, this chapter made the case for greater pragmatism when dealing with consumer class actions by referencing procedural unfairness of an excessive disclosure of consumer data, as opposed to an impossible burden of proof of substantive unfairness for zero-priced offerings. This burden of proof should not be about the excessiveness of zero-pricing (*probation diabolica* for substantive unfairness), but rather about an excessive disclosure of consumer data (procedural unfairness pragmatism for consumer justice).

For consumer justice, this chapter argued that consumer privacy must be inclusive of greater consumer autonomy over economic choices and preferences and greater consumer bargaining of reservation prices with further consumer empowerment in physical and virtual locations.

It has been argued that the sharing and selling of consumer data for algorithmic price discrimination, unequivocally exploits consumer vigilance when shopping online. Endorsing the SSNIP test for differentiated pricing analysis (of consumers who overshare consumer data about their economic interests, preferences, and choices on monopolistic platforms versus less visible and non-affluent consumers who go for traditional shopping rather than digital platforms, even for leisure purposes), this chapter argues that even 5 to 15% price increases are exploitative and inflationary. Therefore, it is time for competition authorities worldwide to prioritise online algorithmic price discrimination and consumer data combinations as pivotal to the enforcement of monopolistic consumer data abuse in digital markets.

Electronic copy available at: https://ssrn.com/abstract=4424967